

being dealt with differently by each manufacturer, but a very common overall approach that is seen is to put a number of light sources (LEDs) on the headset and detecting their position within an environment using stationary light sensors placed around the room (like in the Oculus Rift) or the other way round, with the photoreceptors being on the headset and stationary light sources in the room (Valve's Lighthouse system). Both of these are explored in further detail in the next chapter.

Motion Tracking

When you put on a VR headset, you naturally expect to see your hands in the virtual world as well, moving according to your movements in the physical world. And one of the leading players who have made a significant headway in motion tracking is Leap Motion with their motion tracking accessory. Using infrared sensors to achieve highly detailed motion tracking (down to finger movements), it has been popularly used with the Oculus Rift headsets to allow hand detection. Even though it has been much better than anything we've had so far, upon usage there have been several problems found with the device (like it cannot detect the complete hand when the hand is held perpendicularly to the sensor). The two infrared LED's emit light which is then detected by the cameras on the device. This raw sensor data is passed back to the software on the PC connected to the Leap motion device, where it is mapped into a 3D representation of what the device sees. This is done over time using a series of snapshots which are then programmatically translated into motion detection that allows us to see our hands in virtual reality.

Better devices are coming out in this spectrum like the Oculus Touch or HTC Vive's controllers, but most of them use the principle of photodetection to achieve motion tracking in a manner not too different from Leap Motion.

Special Note:- Eye tracking is being brought into the VR space by FOVE's crowd-funded VR headset. Infrared sensors within the headset detect where the eye is looking, and this allows in-VR elements to react more appropriately to your vision. For example, visual focus can be implemented as opposed to everything being in focus as it usually is on VR headsets.

VR Gloves and Shoes - Inputs or Outputs?

Gloves have been a part of VR for longer than most of us can imagine. Due to the necessity of including the hand in Virtual Reality, this has been one of the approaches adopted to do so. One of the ways in which gloves